

Liberta: A Hybrid Transformer-Linguistic Model for Machine-generated Text Detection

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Abstract

With the rapid advancement of Large Language Models (LLMs) capable of generating text highly resembling human writing, a critical task is to distinguish between human written and machine-generated text. This paper describes our work on the SemEval 2024 Task 8: *Multidomain, Multimodel and Multilingual Machine-Generated Text Detection*. Our method focused on combining RoBERTa embeddings with readability and token-probabilistic features on tackling both the mono- and multilingual track of Subtask A: Human vs Machine Text Classification. On the official test dataset, our system achieved an accuracy of 88.64% in the monolingual and 79.87% in the multilingual track. Our results demonstrate that a hybrid approach can significantly enhance the performance in machine-generated text detection compared to a system that solely relies on a transformer-encoder.

1 Introduction

LLMs have a significant influence on our daily tasks across a wide range of domains, from education and research to content creation and communication. Since their impact on society is significantly shaping how we live and work, the potential risks associated with them are equally substantial. From spreading misinformation (Kreps et al., 2022), gaining unfair advantages in exams to plagiarism (Wahle et al., 2022), and a societal risk when it becomes unclear if communication or information comes from a human or machine, there are many dangers and potentially malicious use cases that make it crucial to develop automated means for detecting machine-generated text. In response, different detection tools have been developed. However, since a great deal of effort is put into developing and improving LLMs, the detection of machine-generated text turned out to be an ongoing arms race where the rapid improvement

of LLMs challenges detectors to deliver accurate results. A striking example for this is the OpenAI classifier, which was released in 2023 and discontinued shortly after due to a low rate of accuracy (Jan Hendrik Kirchner, 2023). The SemEval 2024 Task 8: *Multidomain, Multimodel and Multilingual Machine-Generated Text Detection* addresses this challenge and provides an incentive to find solutions to the detection of machine-generated text across multiple generators (Wang et al., 2024). In the following we describe our solution to the monolingual track of Subtask A: Human vs Machine Text Classification, which is a hybrid model that combines RoBERTa embeddings, readability features and token-level probabilistic features. We also present and discuss the adaption of this hybrid model to the multilingual track of Subtask A, which additionally to English includes Arabic, German and Italian as surprising languages. During the development of our models we adhered to the same conditions as participating teams in the shared task in order to ensure a fair comparison. Our results suggest that a hybrid model that combines those three feature sets can significantly improve classification performance and yield a more robust approach than relying on a PLM alone. While RoBERTa successfully captures semantic nuances of the input text, the engineered features provide complementary information that RoBERTa alone misses.

2 Related Work

For the binary classification task of machine-generated text detection as in Subtask A, there are two main approaches. Zero-shot methods and supervised methods (Wang et al., 2024). Zero-shot methods have the advantage of not relying on curated datasets of human written and machine-generated text. This can be particularly beneficial in the context of the previously mentioned "arms

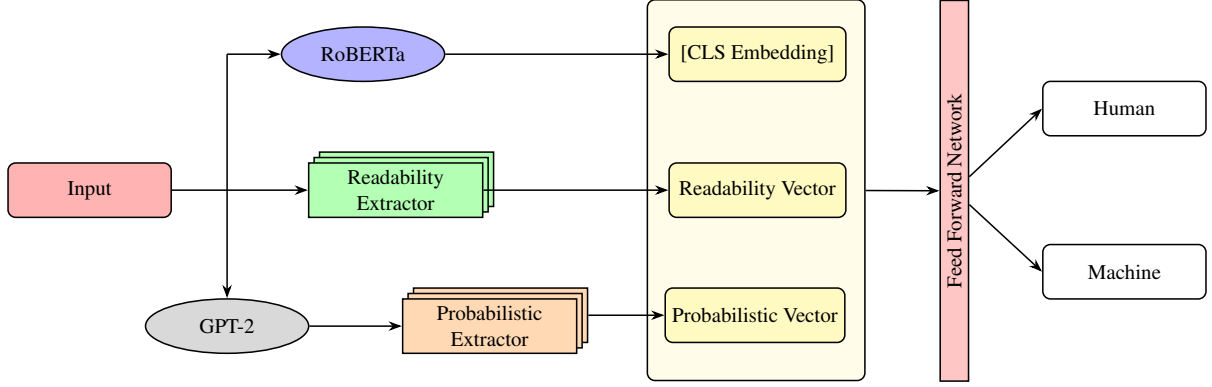


Figure 1: Overview of the proposed Hybrid Model Architecture combining three different feature vectors.

race", where newer LLMs are already far ahead of their predecessors. Although, zero-shot detectors are considered less accurate, innovations like DetectGPT, which only uses log probabilities computed by the model of interest, show that they can accelerate the development of more robust and generalizable methods of machine-generated text detection (Mitchell et al., 2023).

In the shared task, however, most prominently used by the participating teams were supervised training methods. Among those, there are three main groups of methods. The first one is a fine-tuned detection, which uses a pre-trained language model (PLM) and further trains it on the training dataset. Team USTC-BUPT is among the best performing teams at the monolingual track of Subtask A only utilizing PLMs. By adding a gradient reversal layer and a domain classifier that additionally outputs domain labels they achieved an accuracy of 96.10% (Guo et al., 2024).

A second major line focuses on statistical and probabilistic methods. These methods build on the principle that LLMs choose tokens they assign a higher probability to, while humans tend to write in a more "surprising" way. The first ranking Team Genaios employ their model LLMixtic, which mixes token-level probabilistic features from four different Llama-2 models forming a feature vector that is fed into a Transformer Encoder. Their method proved highly effective, achieving an accuracy of 96.88% on the monolingual and 75.67% on the multilingual track (Sarvazyan et al., 2024). Finally, there is a stylistic detection that analyzes linguistic features to distinguish between machine-generated and human written text. By testing a variety of different features as lexical diversity, text statistics and readability features and combining them with CLS embeddings from a transformer en-

coder model, Team PetKaz achieved an accuracy of 90.51% on the monolingual track (Petukhova et al., 2024).

3 Methodology

Figure 1 depicts the architecture of our best performing model. It combines three distinct types of features extracted from an input text. The semantic embeddings of a fine-tuned auto-encoder model, different readability metrics and token-level probabilistic features. This feature vector is passed through a feed-forward neural network for classification.

3.1 Semantic Feature Extraction

We employ RoBERTa-base, fine-tuned on the Machine-generated text classification task. The CLS Token represents the entire input sequence and is placed at the beginning of the input, capturing important semantic and contextual information. For each input we extract the CLS Token embedding, which for RoBERTa-base is 768-dimensional, and use it as the first part of our feature vector.

3.2 Readability Feature Extraction

While today's LLMs generate grammatically correct and coherent text, their output subtly differs from human written text, which we tried to make use of by devoting another part of our feature vector to a detailed linguistic analysis of the input sequence. Team Petkaz at the shared task, whose architecture strongly resembles ours, suggests a range of linguistic features such as text statistics like the number of difficult words, stylistic features, lexical diversity and readability features. Based on our data analysis, we decided to include the latter into our model. Previous works suggest that machine-generated text tends to be more

difficult to read, i.e more complex than human written equivalents. (Georgiou, 2024) for example, reports higher Flesch Reading Ease scores for human texts, which suggest that they are easier to read. Guided by this thesis, we calculate a variety of readability metrics including the following.

Flesch Reading Ease and Flesch-Kincaid Grade Level. Focus on sentence length and syllable count per word.

$$FK = 0.39 \times \frac{\text{total words}}{\text{total sentences}} + 11.8 \times \frac{\text{total syllables}}{\text{total words}} - 15.59$$

The FK Grade-Level was first used by the US army to assess the difficulty of technical manuals. With its weighing factors it outputs a U.S. grade level determining the texts difficulty.

Gunning Fog Index. Assesses sentence complexity by analyzing polysyllabic words.

$$GFI = 0.4 \times \left(\frac{\text{total words}}{\text{total sentences}} + 100 \times \frac{\text{complex words}}{\text{total words}} \right)$$

Words are considered as complex if they consist of three or more syllables. Common suffixes (such as -es, -ed or -ing) are not included. The output is the estimated years of education it requires to read a text of this complexity.

Automated Readability Index. Works like the GFI but instead of counting syllables it counts the number of characters per word and number of words per sentence. and outputs a score that maps to a Grade Level from Kindergarten to College Student.

$$ARI = 4.71 \times \frac{\text{characters}}{\text{words}} + 0.5 \times \frac{\text{total words}}{\text{total sentences}} - 21.43$$

3.3 Token-Probabilistic Feature Extraction

This part of the feature extraction focuses on statistical detection of machine-generated text and follows the core hypotheses of machine-generated text being more predictable than human-written text. While humans tend to write in a more surprising way, LLMs are optimized to choose the next most likely token (Holtzman et al., 2019). To capture these metrics we extract a comprehensive set of features from the token level probability distribution generated by GPT-2.

For a given input text with the tokens $x_1, x_2, \dots, x_n$, we process the sequence with GPT-2. At the i -th position we can extract the probability distribution $P(x_{i+1}|x_1, \dots, x_i)$ over the entire GPT-2 Vocabulary. From this distribution we can derive different features. Here, we used

features inspired by the work of Genaios, the first ranking team in the shared task and got a feature vector including the following metrics, averaged across the entire sequence.

For each position we calculate (i) the log probability of the observed token, measuring how likely the token is, given the tokens already seen. For the feature vector we then use the average observed token probability which essentially describes the perplexity of the input sequence. Perplexity is defined as the exponentiated average negative log-likelihood of a sequence X with t tokens:

$$ppl(X) = \exp \left(-\frac{1}{t} \sum_i \log p_\theta(x_i|x_{<i}) \right)$$

Intuitively, it can be thought of as an evaluation of the model’s uncertainty when predicting the next token. A lower perplexity, and thus a higher average observed log-probability, is a strong indicator of machine-generated text. We further calculate (ii) the probability of the most likely token, which describes the confidence of the model at each step. Finally, we take (iii) the entropy, which measures the uncertainty of the distribution.

$$\beta_i = - \sum_{y \in \mathcal{V}} p_\theta(y|x_{<i}) \log p_\theta(y|x_{<i})$$

A low entropy would indicate that the model is highly confident in a few tokens.

3.4 Feed-forward neural network

The concatenated feature vector is then passed into a feed-forward neural network, where the output layer classifies the text as either human or machine.

4 Experiments

4.1 Dataset and Evaluation

We used the SemEval-2024 Task 8 dataset, which is an extension of the M4 dataset (Wang et al., 2023). The training set includes five domains: Wikipedia, WikiHow, Reddit, arXiv and PeerRead. Machine-generated texts were produced by various LLMs, including davinci-003, ChatGPT, Cohere and Dolly-v2. For the test set Outfox was used as a surprising domain and GPT-4 and BLOOMz are introduced as surprising generators. The labels are roughly balanced with 56,400 machine-generated and 63,351 human-written texts. The multilingual training set additionally contains texts in Chinese, Urdu, Bulgarian and Indonesian, and for the multilingual test set, Arabic, German and Italian are

introduced as unseen languages. Table 1 depicts the dataset statistics. As in the official shared task setting, we adhered to the rules of not using additional training data, and by treating the test set labels as unavailable during the development of our model in order to assure a fair comparison to other teams results. Following the official task description, we employed accuracy as the evaluation metric to determine the quality of our model.

Track	#Train	#Dev	#Test
Monolingual	119,757	5000	34,272
Multilingual	172,417	4000	42,378

Table 1: Dataset statistics for Subtask A

4.2 Experimental Setup

We use RoBERTa-base as the source of the CLS embeddings. For the fine-tuning process we used the baseline framework provided in the task’s description, and slightly modified to support parameter-efficient fine-tuning (PEFT). With limited computing resources at hand, we were searching for ways, to maximize performance, while minimizing computational costs. Low-Rank Adaptation (LoRA) is reported to achieve comparably good or better results than full fine-tuning, despite having fewer trainable parameters. By freezing pre-trained model weights and adding small, trainable matrices to the model’s layers, LoRA drastically reduces the number of trainable parameters. We trained RoBERTa-base for 3 epochs with a learning rate of $2e-4$ and a batch size of 16. For the LoRa configurations, we set the rank of the low-rank matrices r to 64, $lora_alpha$ (scaling factor of the learning rate) at 16 and a dropout of 0.01 to avoid overfitting.

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